

F_U_Bi vs F_U_Bi_i Distinction — Direction, Magnitude, Dimensionality

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PAPER_990: F_U_Bi vs F_U_Bi_i Distinction

Abstract

We formalize the distinction between two complementary UQFF buoyancy quantities that are frequently confused:

Property	$F_{U,\text{Bi}}$	F_{U,Bi_i}
Direction	Inside $\rightarrow$ Outside	Outside $\rightarrow$ Inside (net)
Sign	Always positive	Typically negative
Units	Energy (J)	Acceleration (m/s ²)
Magnitude	$\sim 10^{40}$	$\sim 10^{-2}$
Physical meaning	Vacuum mass transport	Net gravitational acceleration

1. F_U_Bi (Inside-to-Outside)

$$F_{U,\text{Bi}} = \rho_{\text{SCm}} \cdot V \cdot S_{26}^2 \cdot \frac{|U_b|}{|U_g| + |U_b|}$$

This is the total vacuum energy flowing outward through the 26-layer buoyancy structure. It is always positive and cosmologically large because it includes $V_{\text{region}} \sim 10^{48} \text{ m}^3$.

2. F_U_Bi_i (Outside-to-Inside)

$$F_{U,\text{Bi}_i} = U_g + U_m + U_A - U_b + F_n \cdot S_{26} \cdot \Phi \cdot E_{\text{net}}$$

This is the per-particle net acceleration through the 6-layer canonical structure. Negative values indicate net inward pull (gravity dominates buoyancy at the acceleration level).

3. Complementarity

The ratio $|F_{U,\text{Bi}_i}| / F_{U,\text{Bi}}$ gives the fractional acceleration per unit vacuum energy — the "efficiency" of gravitational focusing through the buoyancy medium.

4. Implementation

File: fubi_inside_outside.py, class FUBiDistinctionCalc. CP4 class #574.

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037
 > (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{U_{Bi}}$ jet
 > modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure,
 raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V}{S_{26}^{(3),2}} \cdot \frac{G M}{r_H^2}\right)$$

where:

- L_{Edd} is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3),2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25, \text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M - σ correction (PAPER_1048): The phonon-corrected M - σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
String sector coupling	$[SSq]$	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	Phonon resonance
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37} \text{ J/m}^3$	Fundamental

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Jet power P_{jet}	UQFF phonon-modulated $M_{\text{jet}}(\Gamma)$	Observed $P_{\text{jet}} \sim 10^{43} \text{--} 10^{46} \text{ erg/s}$	Ghisellini et al. (2014)	Within range
$\sin^2 \theta_W$	Embedded in U_{g2} charge coupling	0.2312	PDG 2024	99.6%
Fine structure α	UQFF reproduces via U_{g1} dipole	1/137.036	PDG 2024	99.9%

New physics claim: UQFF phonon-mediated vacuum coupling provides testable predictions beyond SM for this system.

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification ****Sector:**** BH-accretion (active galactic nucleus jet)

§A.2 Lagrangian Density
$$\mathcal{L}_{BH\text{ accretion}} = \sum_{i=1}^{26} \left[U_{g,i} + U_{m,i} + U_{A,i} - U_{b,i} \right] \cdot S_{26}([SSq]) \cdot \Phi_{1.25\text{THz}}(\omega, \Gamma)$$

§A.3 Euler-Lagrange Equation of Motion
$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)}} = 0 \implies F_{U,Bi} = -\nabla U_{\text{eff}} + \Phi \cdot S_{26} \cdot E_{\text{net}}$$

§A.4 Cosmogenesis Linkage Chain PAPER_877 axioms $\rightarrow$ SCm vacuum $\rightarrow$ phonon ω_{SCm} $\rightarrow$ active galactic nucleus jet $\rightarrow$ $F_{U,Bi}$ unified force $\rightarrow$ observational prediction

§B VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)
$$\text{VDS} = \rho_{\text{SCm}} \cdot S_{26} \cdot \Phi_{1.25\text{THz}} / \Phi_0$$
 VDS sub-ratio: 0.167

§B.2 Dipole Vortex Primes (DVP) DVP prime: 73 (resonant)

§B.3 Buoyancy Saturation Harmonics (BSH) BSH timescale: $10^6 M_{\text{BH}}$ yr

§B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.167	Confirmed
κ decay	5×10^{-4}	Confirmed
$[SSq]$	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

- > Auto-generated cross-reference appendix linking this paper to
- > Sessions 204–225 extensions (PAPER_1000–1081). Added by
- > update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)

PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1043	F_U_Bi_i Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation

10 cross-reference(s) identified.

References

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2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Archimedes (~250 BCE). *On Floating Bodies*. (Principle of buoyancy)
4. Churazov, E. et al. (2000). *Evolution of Buoyant Bubbles in M87*. A&A **356**, 788 — arXiv:astro-ph/0004212
5. Fabian, A.C. et al. (2003). *A deep Chandra observation of the Perseus cluster*. MNRAS **344**, L43 — arXiv:astro-ph/0306036 — doi:10.1046/j.1365-8711.2003.06902.x